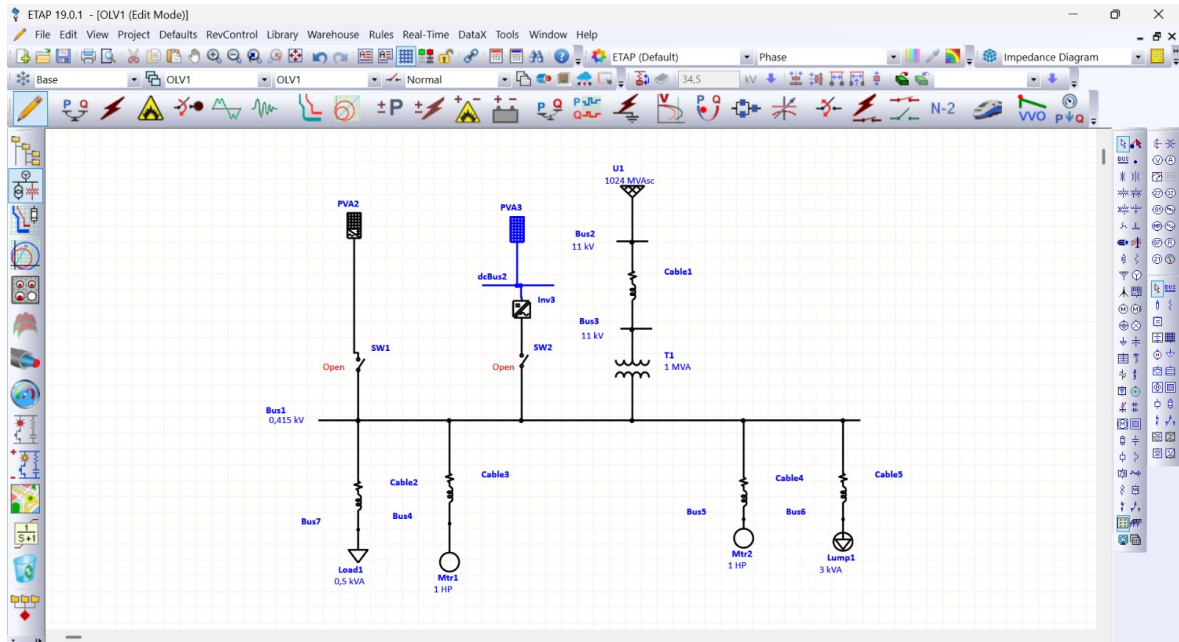


Power System Simulation Report

Grid-Connected Solar PV Integration into an 11 kV Distribution Network



Simulated in ETAP 19.0.1

Software: ETAP (Electrical Transient Analyzer Program)

Designed and Simulated by: [Your Name]

Date: July 2026

1. Project Overview

This project involved designing and simulating a hybrid power distribution network in ETAP that combines conventional utility grid supply with renewable solar PV generation. The model captures how distributed inverter-based solar sources can be integrated into a standard 11 kV medium-voltage distribution system, and how manual switching configurations affect power flow, isolation, and system reliability.

Key System Elements

- Utility grid source (U1) feeding the network through an 11 kV busbar system.
- Two solar PV arrays (PVA2 and PVA3) with inverter-based grid integration (Inv3).
- Power transformer (T1, 1 MVA) stepping down voltage from 11 kV to 0.415 kV.
- Multiple load types — motors, lighting, and general loads — distributed across the low-voltage bus (Bus1).
- Manual disconnect switches (SW1, SW2) for isolation and safety during maintenance.

Objective

The simulation was used to analyze how renewable energy sources can be integrated into a conventional grid-fed distribution system, and to evaluate how different switching configurations impact power flow, voltage levels, and overall network reliability.

2. Single Line Diagram (ETAP)

The figure below shows the complete single line diagram (SLD) as modeled in ETAP, including the utility source, PV arrays, inverter, transformer, buses, cables, switches, and connected loads.

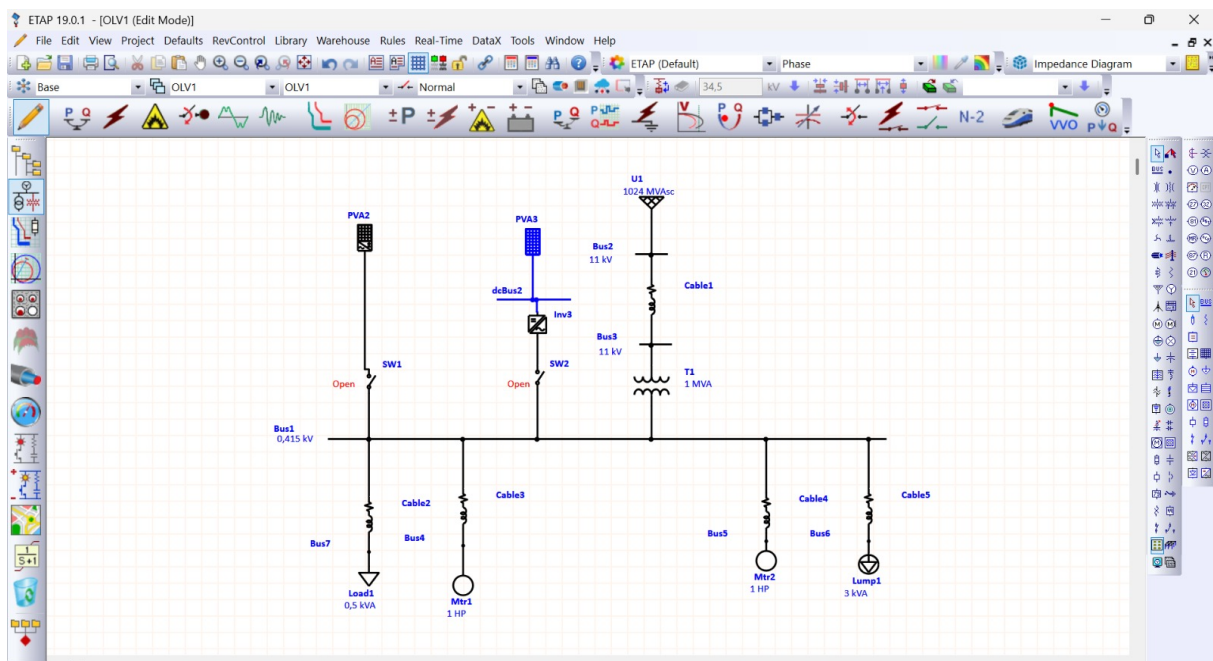


Figure 1 — ETAP single line diagram of the grid-connected solar PV distribution network.

3. Component Summary

Tag	Component	Rating / Voltage	Function
U1	Utility Grid Source	1024 MVA _{sc}	Main grid supply / infinite bus
Bus2 / Bus3	11 kV Busbars	11 kV	Medium-voltage distribution points
Cable1	MV Cable	11 kV	Transfers power between buses
T1	Power Transformer	1 MVA, 11/0.415 kV	Steps down voltage to LV level
Bus1	Main LV Busbar	0.415 kV	Distributes power to loads
PVA2 / PVA3	Solar PV Arrays	DC generation	Convert sunlight to DC electricity
Inv3	Inverter	DC → AC	Converts PV DC output to AC for grid injection
SW1 / SW2	Disconnect Switches	Manual	Isolation for safety / maintenance
Load1	General Load	0.5 kVA	Small lighting / general load
Mtr1 / Mtr2	Induction Motors	1 HP each	Motor-driven equipment (pumps, fans)
Lump1	Lumped Load	3 kVA	Larger lighting / equipment load

4. Real-World Interpretation of the Diagram

To make the schematic accessible to a non-specialist audience, each element of the ETAP model was mapped to its real-world physical equivalent, as summarized in the infographic below.

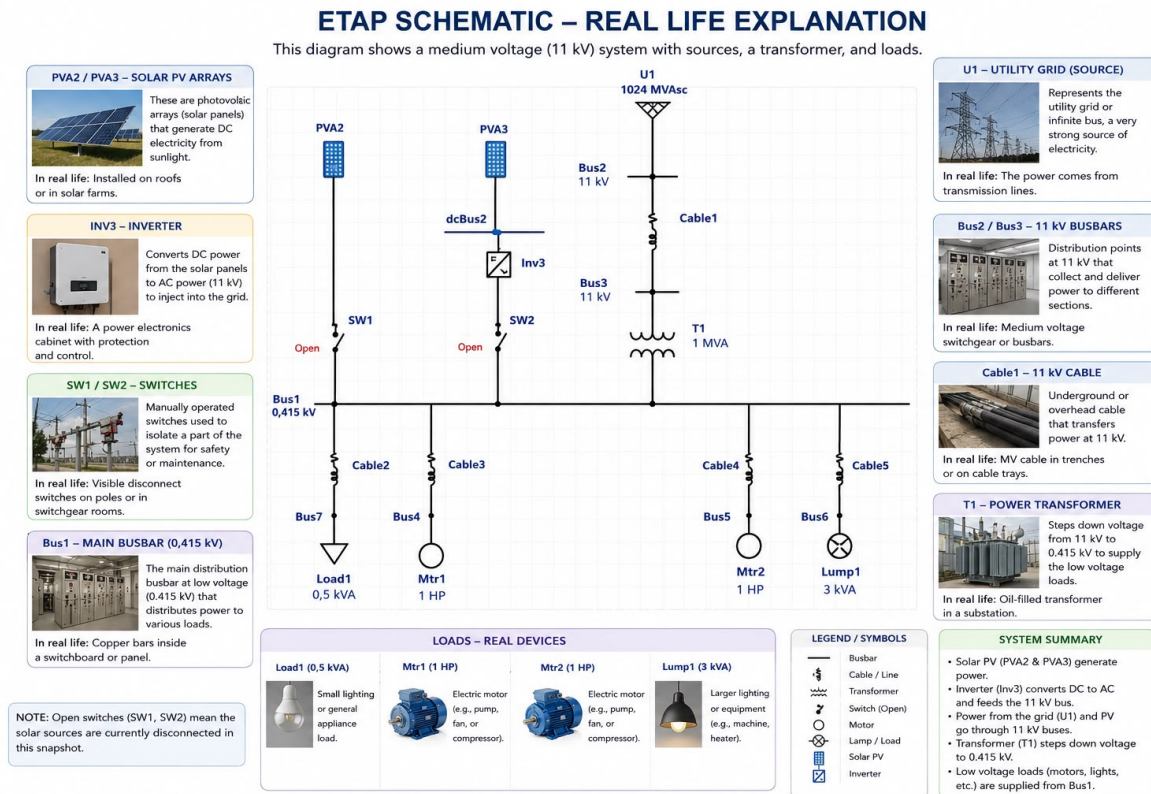


Figure 2 — Real-life explanation of each component shown in the ETAP schematic.

5. Observations & System Summary

- Solar PV arrays PVA2 and PVA3 generate DC power, which is converted to AC by inverter Inv3 before being fed into the 11 kV bus.
- Power from both the utility grid (U1) and the PV sources is combined and distributed through the 11 kV buses (Bus2, Bus3).
- Transformer T1 steps the voltage down from 11 kV to 0.415 kV to supply the low-voltage distribution bus (Bus1).
- Low-voltage loads — including motors (Mtr1, Mtr2), general load (Load1), and lumped load (Lump1) — are all supplied from Bus1.
- Switches SW1 and SW2 allow the PV branches to be isolated independently for maintenance without interrupting supply to the rest of the network. In the analyzed snapshot, both switches were open, meaning the solar sources were disconnected from the network.

6. Conclusion

This exercise demonstrates a practical approach to modeling renewable energy integration within a conventional utility-fed distribution network using ETAP. The model provides a foundation for further studies such as load flow analysis, short-circuit analysis, and protection coordination under different solar generation and switching scenarios.